



DNnet: A lightweight network for real-time 4K underwater image enhancement using dynamic range and average normalization

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ABSTRACT

In recent years, neural network-based methods have emerged in underwater image enhancement (UIE). Although researchers have made considerable progress in this field, applying state-of-the-art algorithms requires significant computing resources for real-time applications, making it impossible to deploy on underwater mobile devices. To address this issue, we propose the pixelunshuffle-bottleneck-pixelshuffle structure, which utilizes more parameters to improve the parallel computational efficiency and achieve higher computing efficiency. In addition, we propose fast average normalization (FAN) with a channel dynamic range (CDR) module to ensure the enhancement quality during high-speed inference. FAN uses average pooling to reduce overenhancement or underenhancement risks on channel-wise normalization. CDR can utilize dynamic range to further improve the enhanced image quality given by FAN. Moreover, we introduce a color column loss to evaluate the local color distribution between the generated image and ground truth to avoid the undersaturation problem. Our proposed method¹ outperforms existing state-of-the-art methods, with a peak signal-to-noise ratio of 26.335 and a structural similarity index measure of 0.91 on the UIEB dataset. Furthermore, our method can achieve real-time enhancement of 4K images on edge devices up to 78FPS.

1. Introduction

Underwater images often exhibit low contrast and blue-green tones owing to significant attenuation and scattering effects, leading to color distortion and blurred details in underwater images (Chang, Li, Ren, & Chen, 2024; Hao, Longxiang, Xiangchun, Fei, & Jiawei, 2023). To meet the growing demand for marine exploration, resource development, and autonomous underwater imaging and navigation, various underwater image enhancement (UIE) algorithms have been developed (Ancuti, Ancuti, De Vleeschouwer, & Bekaert, 2017; Muniraj & Dhandapani, 2021; Raveendran, Patil, & Birajdar, 2021; Yeh, Lin, Kang, Huang, & Wang, 2021; Zhuang, Li, & Wu, 2021).

UIE algorithms can be broadly categorized into learning-free and learning-based methods (Guan et al., 2023). Learning-free methods (Hitam, Awalludin, Yussof, & Bachok, 2013; Jha & Bhandari, 2024; Liang, Wang, Ding, Mi, & Fu, 2021; Zhou, Pang, Zhang, & Zhang, 2023), being fast on CPUs, offer efficient solutions for specific underwater scenarios. Learning-based methods, though computationally demanding, excel in handling more complex underwater conditions.

With the continuous improvement of UIE datasets (Li et al., 2019; Peng, Zhu, & Bian, 2023) and the development of model architectures,

learning-based UIE algorithms have made great progress in mitigating issues such as color distortion and low contrast. However, the high computational requirements of these advanced algorithms (Naik, Swarnakar, & Mittal, 2021) pose challenges, especially for mobile deployment (Zhang, Zhao, An, Li, & Zhao, 2024).

Previous solutions for fast UIE, such as gradually shrinking features (Islam, Xia, & Sattar, 2020), reducing channel dimension (Li, Anwar, & Porikli, 2020; Naik et al., 2021), and adopting computationally efficient convolutions (Zhang et al., 2024), speed up inference mainly by lowering computational complexity. However, the models designed following these strategies are still not sufficient (Lee, Hwang, Lee, Bae, & Park, 2019) for real-time processing of high-resolution images up to 4K, which are now commonly captured by modern underwater vehicles (Xie et al., 2024) (Fig. 2).

By contrast, inspired by works in lightweight super-resolution models (Li, Fang, Mei, & Zhang, 2018; Liang, Cao, et al., 2021), we improve the parallel computational efficiency by utilizing more parameters to process the same amount of calculation through pixelunshuffle-bottleneck-pixelshuffle (PBP) structure to achieve superior inference speed. Our proposed model using this structure can eventually achieve

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¹ Our code will be available at <https://github.com/Tian-Yu-CAO>.

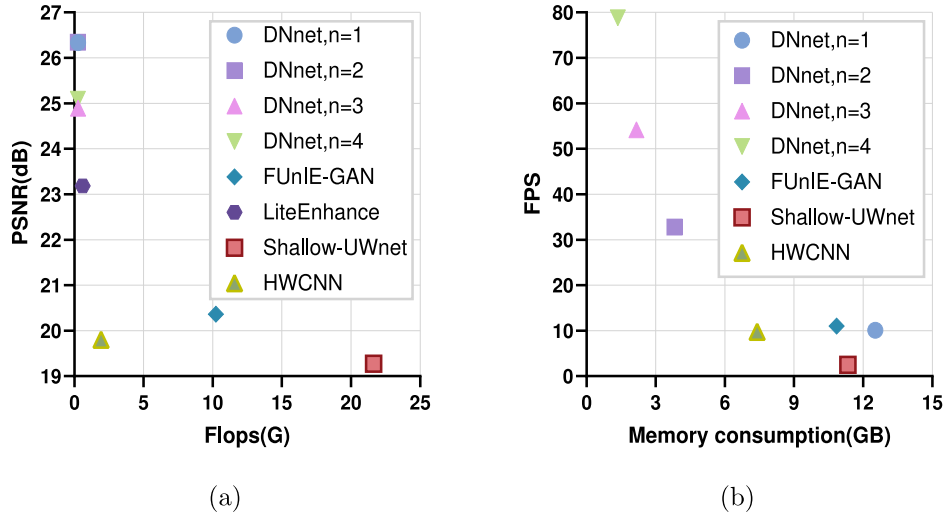


Fig. 1. PSNR vs. floating point operations per second (FLOPs). (b) FPS and memory consumption when tested on 4K images. The proposed DNnet can achieve the fastest inference speed while achieving state-of-the-art performance. All models were tested using a 4090 laptop GPU.

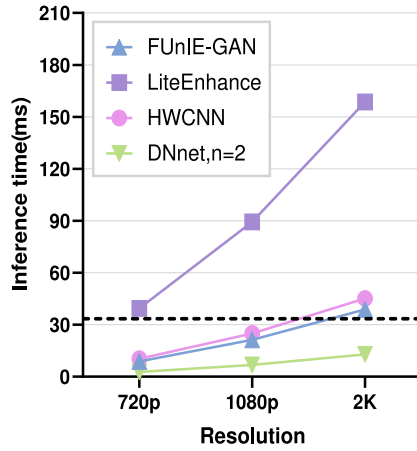


Fig. 2. Comparison of inference time between mainstream lightweight UIE models (Islam et al., 2020; Li et al., 2020; Zhang et al., 2024) and our proposed DNnet ($n = 2$). The vertical axis represents the time required to infer an image on a 4090Laptop GPU when the model is stable. The horizontal axis represents the resolution of the image input to the model. Nodes below the dotted line indicate that a model meets the real-time (30FPS) requirements. Our proposed DNnet ($n = 2$) achieves real-time inference speed at 720p 1080p and 2K resolutions.

real-time 4K inference (Fig. 1).

Despite the advantages of improved speed and efficiency, balancing high-quality output with fast inference remains a challenge. Recent learning-based methods (Guo et al., 2023; Islam et al., 2020; Park & Eom, 2024), particularly those incorporating normalization, have demonstrated notable improvements in image quality. For example, FUnIE-GAN (Islam et al., 2020) enhanced overall image quality by normalizing the results of the neural network. ASANet (Park & Eom, 2024) incorporated a normalization module with maximum- and minimum-value estimation and channel attention to achieve better visual quality. However, the normalization in these methods did not account for overly bright or dark areas caused by irregular illumination and other factors. These limitations result in overenhancement or underenhancement. Also, complex normalization designs lead to low operating efficiency.

To overcome these limitations and simplify design, we introduce a fast average normalization module (FAN) that uses average pooling to reduce computations and the impact of uneven illumination. On the other hand, previous methods combined normalization mainly by

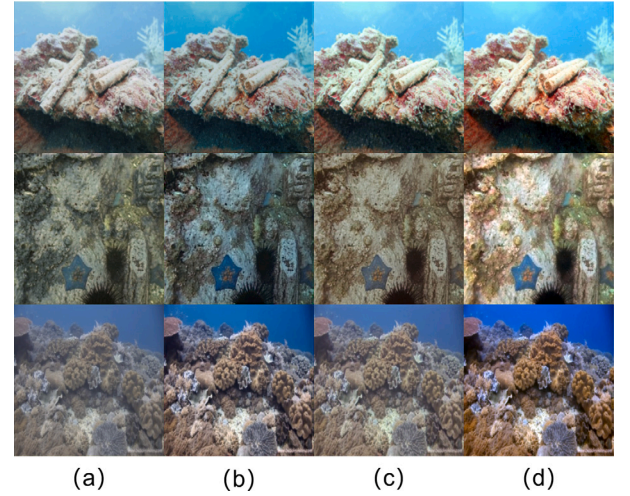


Fig. 3. Demonstrations of enhancement results. (a) Shallow-UWnet, (b) LiteEnhanceNet, (c) HWCNN, and (d) Proposed method.

directly inserting the normalization module within the model (Guo et al., 2023) or at the end (Islam et al., 2020). By contrast, inspired by Fusion (Ancuti et al., 2017), we develop a channel dynamic range (CDR) module that processes image dynamic range information to adjust the relationship between normalization and the model by observing the enhanced results of normalization. Additionally, we present a dynamic range and normalization framework, DNnet, which combines BPB, CDR, and FAN to achieve high-quality and efficient inference.

In addition to the above improvements, we believe that the loss function is equally important for UIE. Many learning-based methods (Li et al., 2020; Naik et al., 2021; Zhang et al., 2024) still rely on L1 and L2 losses to measure the gap between the ground truth (GT) and enhanced results. However, our observations indicate that models trained with pixel-level losses often suffer from insufficient saturation (Fig. 3). Because the local color distribution between the GT and undersaturated image is significantly different, we propose a local color distribution-sensitive loss, named color column loss, that focuses on this difference to improve the accuracy of saturation.

In summary, we design DNnet for the real-time high-resolution UIE tasks. Particularly, the contributions of our work are as follows:

- (1) The proposed DNnet comprises lightweight PBP structures, which improve the parallel computational efficiency by utilizing more parameters in exchange for inference speed. Consequently, our method can enhance 4K images in real time on multiple edge devices.
- (2) We propose the FAN module, which uses average pooling to reduce the influence of overly bright or dark areas caused by irregular illumination to improve robustness.
- (3) We introduce the CDR module to coordinate FAN and the PBP and improve the quality of the enhanced images.
- (4) We introduce the color column loss, which focuses on local color distribution, to address the undersaturation problem caused by training with pixel-level loss.

The remainder of the article is organized as follows: Section 2 briefly introduces the UIE works related to our method. Section 3 describes the implementation details of each proposed module, as well as loss-related settings. Section 4 discusses the various experiments for mass analysis. Section 5 summarizes the content of our study.

2. Related works

2.1. Lightweight UIE methods

There are several methods to implement deep learning-based algorithms on resource-constrained devices. For instance, designing lightweight convolutional neural networks (CNNs) is one of the main methods. This method aims to explore the sparse hierarchical structure of CNNs without significantly reducing the network accuracy (Chen, Li, Han, Ren, & Yang, 2023). Introducing specific convolutional paradigms, such as group convolution (Krizhevsky, Sutskever, & Hinton, 2012), separable convolution (Howard et al., 2017), and dilated convolution (Yu & Koltun, 2015), can reduce the number of parameters and computational complexity. In addition, to improve network training speed, AlexNet used group convolution to run in parallel on two GPUs. Depthwise separable (DS) convolution, such as depthwise and pixelwise convolution, was widely used in MobileNet (Howard et al., 2017) designed for mobile and embedded devices to improve memory efficiency.

In the UIE field, ADMNNet (Yan, Qin, Wang, Wang, & Fu, 2022) contained DS convolution and a lightweight attention module to accelerate the model. Based on reducing the number of model parameters and calculations, DS convolution was used as the main structure in the LiteEnhanceNet (Zhang et al., 2024) except for the first and last layers. However, in practice, although depthwise convolution and pixelwise convolution have low calculation density, their memory access frequency is higher than that of general convolution when the calculation amount is the same (Zhang, Zhou, Lin, & Sun, 2018), negatively affecting the throughput of the model.

The bottleneck structure, which can be observed in ResNet (He, Zhang, Ren, & Sun, 2016) and SE module (Hu, Shen, & Sun, 2018), improves the operating efficiency of the model by reducing channels. Inspired by the bottleneck, the dense connection in HWCNN (Li et al., 2020) was followed by a channel compression operation so that the channels of the entire model were maintained at a low level. Zhang et al. (2023) designed a multiscale spatial attention mechanism in the bottleneck layer to help the model better understand the geometric structure and semantic information of an image.

For parameterless and fast downsampling and upsampling methods, pixelshuffle and pixelunshuffle (Shi et al., 2016) are widely used in image super-resolution, achieving excellent results. The AutoEnhancer (Tang, Iwaguchi, Kawasaki, Sagawa, & Furukawa, 2022) combining neural architecture search (Zoph & Le, 2016) with pixelshuffle and pixelunshuffle achieved a good balance between model scale and UIE performance.

To summarize the lightweight techniques of existing UIE methods, a comparative analysis of related studies was conducted, as shown in Table 1.

Table 1

Comparative analysis of related lightweight UIE methods.

Methods	Main techniques
HWCNN (Li et al., 2020)	Channel compression
FUnIE-GAN (Islam et al., 2020)	U-Net structure
Shallow-UWnet (Naik et al., 2021)	Reducing layers
AutoEnhancer (Tang et al., 2022)	Neural architecture search
HAAM-GAN (Zhang et al., 2023)	Bottleneck structure
LiteEnhance (Zhang et al., 2024)	DS convolution with SE module

Table 2

Comparative analysis of learning-based UIE methods with learning-free methods.

Methods	Main techniques
FUnIE-GAN (Islam et al., 2020)	Channel-wise normalization
UD-net (Zheng & Luo, 2022)	CNN with CLAHE
PUIE (Fu et al., 2022)	Probabilistic AdaIN
NU ₂ Net (Guo et al., 2023)	Normalization tail
ASANnet (Park & Eom, 2024)	Normalization with attention

2.2. Learning-based UIE methods with learning-free methods

Learning-free and learning-based methods have their advantages. Compared with learning-free methods, neural networks exhibit good non-linear learning capabilities (Salman et al., 2016) that can adapt to different underwater environments based on training data. Meanwhile, machine learning-based algorithms generally use GPUs to accelerate (Pandey et al., 2022). These features can improve the usability and robustness of learning-based UIE algorithms. Compared with learning-based methods, learning-free methods can adjust the overall color distribution of underwater images (Park & Eom, 2024) using less parameters that do not require training datasets. Moreover, enhancing specific environments can be achieved by fusing multiple learning-free methods.

More recently, methods that combined learning-free and learning-based approaches have demonstrated additional advantages. For instance, in the work of Zheng and Luo (2022), a fusion algorithm was constructed, which involved a color balance algorithm, a CNN-based defogging algorithm, and contrast limited adaptive histogram equalization (CLAHE) (Pizer et al., 1987). In addition, the color balance algorithm could eliminate the color deviation caused by the attenuation of light in the water, and the defogging algorithm and CLAHE were used to improve image contrast. PUIE (Fu, Wang, Huang, Ding, & Ma, 2022) introduces a probabilistic adaptive instance normalization (PAdaIN) approach that combines conditional variational autoencoders (CVAE) (Rezende, Mohamed, & Wierstra, 2014) with adaptive instance normalization (AdaIN) (Huang & Belongie, 2017) to generate diverse and flexible enhancement distributions. Moreover, Park and Eom (2024) focused on combining image standardization and image normalization with the learning-based UIE algorithms by incorporating the results of two squeeze-and-excitation blocks (Hu et al., 2018) as the boundary value of normalization. Table 2 presents the comparative analysis of learning-based UIE methods with learning-free methods.

2.3. Gaps in existing research

The methods (Islam et al., 2020; Li et al., 2020; Naik et al., 2021; Zhang et al., 2023, 2024) described in Section 2.1 reduce the calculation amounts to speed up inference. However, they can hardly meet the requirements of real-time enhancement at 4K resolution (Lee et al., 2019). Unlike the methods mentioned in Section 2.1, our proposed PBP structure can significantly improve parallel computational efficiency by utilizing more parameters in exchange for inference speed (Section 3.1).

Section 2.2 presents various UIE methods (Fu et al., 2022; Guo et al., 2023; Islam et al., 2020; Park & Eom, 2024; Zheng & Luo, 2022) combined with normalization. Different from adding training parameters to improve the robustness of normalization (Park & Eom,

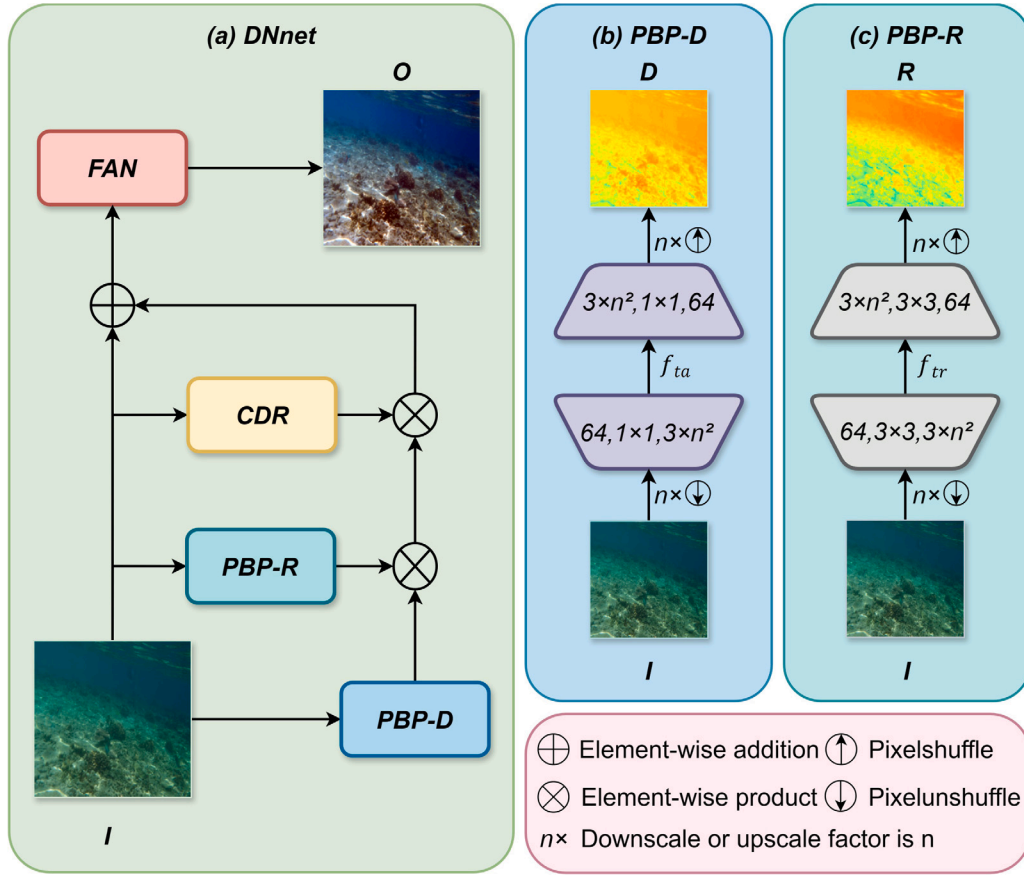


Fig. 4. (a) Overall structure of the proposed DNnet, (b) PBP-D, and (c) PBP-R.

2024), we use average pooling to reduce the risk caused by irregular reflection (Section 3.3). In addition, we introduce the CDR module to adjust the model enhancement strength according to the correlation between dynamic range and normalized image quality (Section 3.2).

3. Methodology

Fig. 4(a) presents the overall structure of our model; it consists of four parts: FAN, CDR, PBP-distance (PBP-D), and PBP-relu (PBP-R). For all convolutional layers in our model, the stride is set to 1 and padding is 1 as well. “ $64, 3 \times 3, 3 \times n^2$ ” indicates that the output channels of the convolution layer are 64, the kernel size is 3×3 , and the number of input channels is $3 \times n^2$.

The calculation formula (Treibitz & Schechner, 2009) for each pixel $L(x, y)$ of the image without the influence of water in an ideal environment is shown in Eq. (1), where $I(x, y)$ represents the pixel located at coordinates (x, y) of the underwater image I , $A_\infty[1 - t(x, y)]$ represents the influence caused by scattering, and $t(x, y)$ represents the influence caused by absorption.

$$L(x, y) = \frac{I(x, y) - A_\infty[1 - t(x, y)]}{t(x, y)} \quad (1)$$

Due to the involvement of multiple unobservable variables, such as distance, brightness and concentration, in the calculation of scattering and absorption, it is extremely difficult to directly restore images based on Eq. (1). If we linearly transform Eq. (1), we can get Eq. (2). Through Eq. (2), the underwater image enhancement process can be simplified to the process of predicting the compensation of all channels based on the raw underwater image I .

$$L(x, y) = I(x, y) + \frac{I(x, y)[1 - t(x, y)] - A_\infty[1 - t(x, y)]}{t(x, y)} \quad (2)$$

Influenced by the idea (Ancuti et al., 2017) of using the distance to the green channel as a cue to compensate the red channel, we introduce full channel compensation strategy which utilizes a neural network to learn how to use the distance information obtained from another neural network as a prompt to perform compensation operations on underwater images.

Assuming that our model processes the underwater image I to obtain the enhanced image O , the processing progress of DNnet can be expressed as follows:

$$O = FAN(C \otimes R \otimes D + I) \quad (3)$$

where D is the feature map or output given by PBP-D, R is the feature map output of PBP-R, C is the weight output given by CDR, and $FAN()$ is the functional representation of the FAN module. Different from the white balance method with compensation in Fusion (Ancuti et al., 2017), we replace the compensation part by $C \otimes R \otimes D$. Because the white balance method cannot support backpropagation, we introduce FAN instead of the white balance method of Fusion for color balance and to reduce overenhancement or underenhancement risks in normalization.

3.1. PBP

3.1.1. PBP structure

According to Lee et al. (2019), reducing FLOPs of the network architectures for speed is based on the condition that every floating point operation runs at the same speed in a device. However, for GPU, the processing speeds of various structures are different, which makes speed reduction measures not always effective. To design an efficient UIE model for parallel computing, we propose the PBP structure (Fig. 5) mainly to improve the computation efficiency. Unlike the conventional

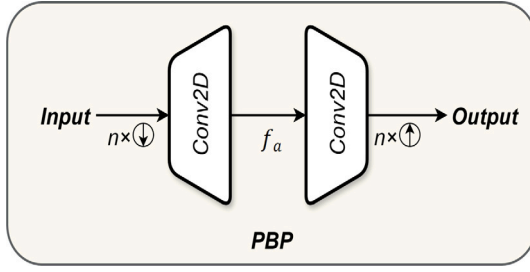


Fig. 5. Overall structure of the PBP. The number of channels between two convolutional layers is fixed to 64.

Table 3

MACs, parameters, computation efficiency, memory usage, and feature map size of PBPs with different n when inferring 4K images.

PBPs	MACs	Params	Computation efficiency (GFlops/ms)	Memory usage (GB)	Feature map size
	(G)	(k)			
$n = 1$	28.67	3.46	1.31	2.50	3840×2160
$n = 2$	28.67	13.82	3.93	1.24	1920×1080
$n = 3$	28.67	31.10	5.68	0.97	1260×720
$n = 4$	28.67	55.30	6.43	0.87	960×540

bottleneck strategy (He et al., 2016), which allows changeable channels according to the expansion ratio, the proposed PBP structure includes a paired pixelshuffle and pixelunshuffle operation to reduce the feature map size while forcing the channels (computational cost) of the inverse bottleneck to remain unchanged. This strategy can accelerate the inference speed by amplifying the number of parameters n^2 times (Table 3). Although this strategy will lead to a sudden increase in parameters, the additional memory usage caused by the increased parameters will not affect the inference speed when the memory of model parameters is much smaller than the memory occupied by feature maps. Because the parameters of the proposed PBP module are quite limited (much smaller than the number of feature maps), our PBP structure can efficiently reduce GPU inference time. Because the PBP structure is composed of general convolutions, the computational cost of the layer near the head and rear in PBP can be calculated as follows (Howard et al., 2017):

$$D_K^2 \times (N \times n^2) \times H \times (D_F^2 \times 1/n^2) \quad (4)$$

where the computational cost depends on the number of input channels of PBP $M \times n^2$ and output channels of PBP $N \times n^2$, the number of output channels of head convolutional layer or input channels of rear convolutional layer H , the kernel size D_K^2 , and the feature map size ($D_F^2 \times 1/n^2$), and n is the upscale factor in pixelshuffle or downscale factor in pixelunshuffle (Shi et al., 2016). In our method, the H is fixed to 64. As shown in Table 3, the feature map size in PBP structure gradually decreases with the increase of n . Notably, PBP with different parameters n has the same computational cost because $n^2 \times 1/n^2 = 1$.

The computational efficiency of the model is significantly improved by keeping calculation complexity unchanged and utilizing more parameters when n increases (Table 3). For more experimental setup details please refer to Section 4.1.

3.1.2. PBP-D and PBP-R

Differing from compensation in Fusion, which was based on the distance between the green channel and other channels, we propose PBP-D (D refers to distance) based on PBP, which adopts Eq. (5) to allow the model to learn the distance relationship between all channels. In addition, PBP-D uses 1×1 convolution kernels to minimize the computational cost. Inspired by the recent work of Ma, Dai, Bai, Wang, and Fu (2024), using element-wise multiplication can significantly amplify the feature dimensions without incurring additional computational

overhead within a single layer. Therefore, we introduce PBP-R (R refers to ReLU), which adopts Eq. (6), to element-wise multiply with PBP-D for enriching the non-linear capabilities of our model (Fig. 4(b)(c)). Another advantage of element-wise operation between PBP-R and PBP-D is that the bottleneck structures in both PBP-D and PBP-R can obtain a large compression ratio between the input and output channels. Section 4.5.1 discusses the impact of n .

$$f_{fa}(x) = |\tanh(x)| \quad (5)$$

$$f_{fr}(x) = \max(0, \tanh(x)) \quad (6)$$

3.2. FAN

Recent works by Park and Eom (2024), Islam et al. (2020), and Guo et al. (2023) have indicated that applying normalization to neural networks is more effective in enhancing underwater images. Owing to the scattering and absorption phenomena in underwater environments, some overly bright or dark areas can be generated in underwater images, thereby affecting the robustness of normalization. Therefore, we use average pooling to reduce overenhancement or underenhancement risks and improve the normalization performance. In addition, we propose a FAN module (Fig. 6) to normalize the input X . This module first performs average pooling on X to get X_A to reduce the influence of overly bright or dark areas. The kernel size and stride of average pool are 8 when resolution is 256×256 . When using images of other resolutions for inference, we linearly adjust the kernel size and stride according to Eq. (8), where H_F represents the kernel size and stride. H, W represent the height and width of the input image. Subsequently, FAN filters out the maximum and minimum values in each channel of X_A . Using the maximum and minimum values, we normalize the input X using Eq. (7) to get the output O . $\max_{i,j}()$ or $\min_{i,j}()$ means to get the maximum or minimum values of each channel. ϵ_n is a small number that prevents the denominator from being zero. Moreover, to validate the efficiency of the proposed normalization method, we conduct an ablation study on various types of normalization, as discussed in Section 4.5.3.

$$O = \frac{X - \min_{i,j}(X_A(i, j))}{\max_{i,j}(X_A(i, j)) - \min_{i,j}(X_A(i, j)) + \epsilon_n} \quad (7)$$

$$H_F = 8 + \frac{\min(H, W) - 256}{2160 - 256} \quad (8)$$

3.3. CDR

Many works have mentioned that the color balance (or white color balance) strategy achieves plausible enhancement results under low degradation (Ancuti et al., 2017; Park & Eom, 2024). By contrast, the learning-based method is more suitable to enhance images under a high degree of degradation (Zhang et al., 2024). To balance FAN and the product of PBP-R and PBP-D, we need a factor to determine the degradation level of the raw underwater image. We found that the dynamic range (DR) (Aydin, Mantiuk, Myszkowski, & Seidel, 2008; Chaurasiya & Ramakrishnan, 2013), which is widely used in photography to analyze images, can also evaluate the quality of raw underwater images.

In our method, we use Eq. (9) to calculate the dynamic range DR. The maximum and minimum pixel values in Eq. (9) are derived from images I_A produced by average pooling with large kernel. Due to the usage of average pooling, these values incorporate contextual information from neighboring pixels, thereby providing a broader representation of the image. Our study (Section 4.5.3) demonstrates that the dynamic range obtained through this average pooling process has a meaningful correlation with the degradation level of underwater images. Typically, a highly degraded underwater image should have a small DR. For our framework, we expect the products of PBP-R and

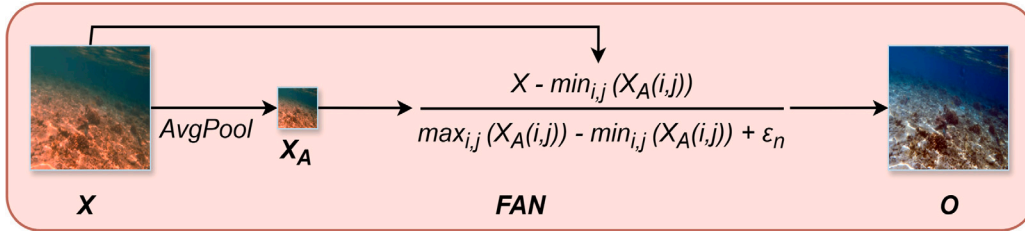


Fig. 6. Overall structure of the FAN.

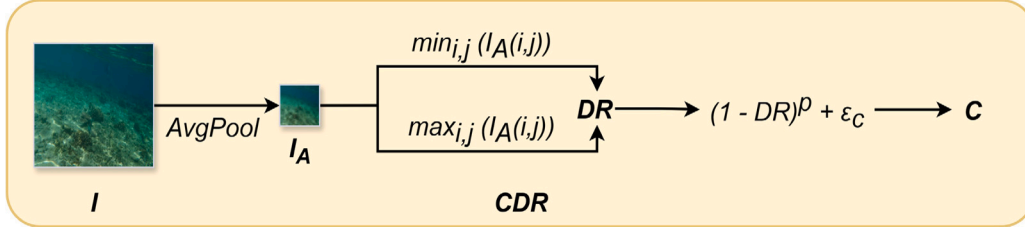


Fig. 7. Overall structure of the CDR.

PBP-D to have an enormous weight when the input image is highly degraded. Thus, we design the channel matrix C , which should have a negative correlation of DR , to control the ratio of how much the PBP branch can affect the final output. Because we design C to directly multiply with PBP-D and PBP-R, we add a positive constant ε_c to Eq. (10) to ensure PBP-D and PBP-R can influence the enhancement result when $1 - DR = 0$. Moreover, we introduce the CDR module (Fig. 7) to obtain C . CDR downsamples the raw image I to get I_A . The kernel size and stride of average pool are 6 when resolution is 256×256 . When using images of other resolutions for inference, we linearly adjust the kernel size and stride according to Eq. (11), where H_C represents the kernel size and stride, H, W represent the height and width of the input image. After average pooling, CDR generates the weights C using Eq. (10) to coordinate FAN and the product of PBP-R and PBP-D.

$$DR = \max_{i,j}(I_A(i,j)) - \min_{i,j}(I_A(i,j)) \quad (9)$$

On this basis, we further propose C as follows:

$$C = (1 - DR)^p + \varepsilon_c \quad (10)$$

$$H_C = 6 + \frac{\min(H, W) - 256}{2160 - 256} \quad (11)$$

In Eq. (10), p is a polynomial factor and is set to 5. Section 4.5.2 discusses the impact of different p . Also, Section 4.5.2 explains the significance and reasons of setting $p = 5$. To our experience, ε_c is set to 0.05 to retain the basic adjustment ability of PBP-D and PBP-R.

3.4. Loss function

The loss function used in this study comprises two parts.

Color column loss: Owing to the uncertainty of pixel-level differences (see fluctuations in Fig. 8(a)), neural networks prefer neutral colors to generate underenhanced results, leading to undersaturation (Fig. 3). To address this issue, we propose the color column loss, which compares local color distribution similarity within a set range between GT images and generated images. The color column loss uses local bins (Fig. 8(b)) to help the model overcome the side effects of fluctuating color distribution. According to our observation on UIEB dataset (Li et al., 2019), we found a large variation between the GT and enhanced image of 2, 3, 6, and 7 bins (Fig. 8(c)) when the image tends to be gray. Thus, we use Eq. (12) to calculate the loss for only bins (Fig. 8(c)) 2,

3, 6, and 7 to address the undersaturation problem. The color column loss is calculated as follows:

$$L_{color\ column} = \frac{\gamma}{T} \sum_{\omega} \left(\frac{\beta(x)_{\omega}}{\beta(x)_{\omega} + \varepsilon_s} - \frac{\beta(y)_{\omega}}{\beta(y)_{\omega} + \varepsilon_s} \right)^2 \quad (12)$$

where γ is the scaling coefficient used to adjust the scale of the loss, T is the total number of pixels, Ω denotes all selected bins (2, 3, 6, 7), and ω is a specific bin of Ω . $\beta(x)_{\omega}$ represents the extracted pixels of the generated image x in bin ω . $\beta(y)_{\omega}$ represents the extracted pixels of GT y in bin ω . ε_s enables color column loss to affect the trainable parameters through backpropagation. $\beta(x)$ is used to extract pixels in x from each bin of Ω . Eq. (13) demonstrates the process of $\beta(x)$ where Δ is the reciprocal of the number of bins. Substitute y for x in Eq. (13) to calculate $\beta(y)$.

$$\beta(x) = [\max(0, -\max(0, -x + \Delta \times (\omega - 1)) + \Delta)]_{\omega}^{\Omega} \quad (13)$$

Structural similarity index measure (SSIM) loss: We use SSIM loss (Wang, Bovik, Sheikh, & Simoncelli, 2004) to measure the structural similarity between the generated image and GT. SSIM loss is expressed as follows:

$$L_{SSIM} = 1 - \frac{1}{N} \sum_{i=1}^N SSIM(x_i, y_i) \quad (14)$$

where SSIM represents structural similarity, and SSIM can evaluate the brightness, contrast, and structural similarity at the image block level. SSIM is shown in Eq. (18).

The final loss function is shown in Eq. (15). Empirically, we set λ_1 to 1 and λ_2 to 0.1. Section 4.5.4 discusses the loss functions.

$$L_{Total} = \lambda_1 L_{SSIM} + \lambda_2 L_{color\ column} \quad (15)$$

3.5. Inference at different resolutions

In DNnet, PBP-D and PBP-R have trainable parameters, while FAN and CDR do not. To enable DNnet, which is trained on low-resolution data, to handle images of higher resolution (such as 4K), we propose an inference process which is shown in Fig. 9. Due to the global adjustment capabilities of FAN and CDR, their kernels and strides need to be readjusted according to Eqs. (8) and (11) during inference. In contrast, PBP-D and PBP-R only allow for local adjustments, so we directly use the weights trained at lower resolution for higher resolution inference.

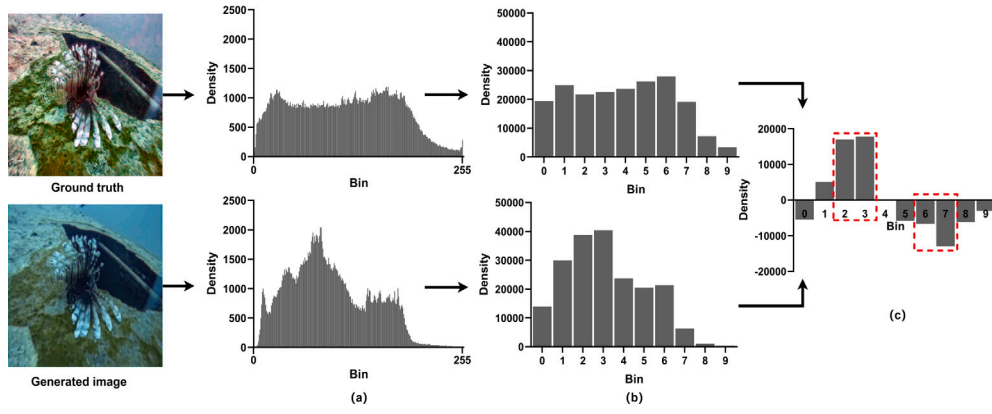


Fig. 8. Details of color column loss function. (a) Full color distribution of GT and generated image. (b) Adapting the full color distribution of GT and generated image into 10 bins. (c) Differences between GT and generated image in the form of 10 bins.

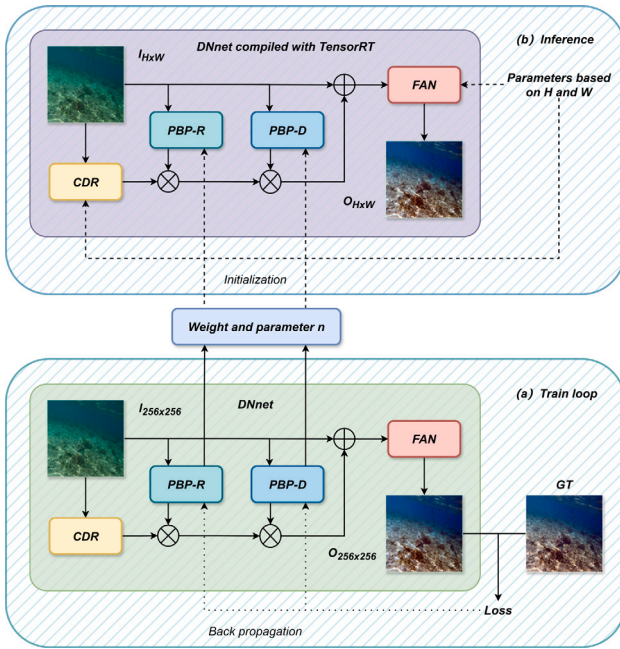


Fig. 9. DNnet uses low resolution images for training and achieves high-resolution inference by loading weights or parameters to the model using different strategies. (a) The train loop of DNnet, (b) The inference of DNnet. The dashed line represents the initialization process, and the dotted line represents the back propagation process.

4. Experiments and analysis

This section introduces the details of the experiment and the datasets, discusses the detailed qualitative and quantitative analyses and ablation studies, and presents the analyses of the efficiency of our method.

4.1. Experimental details

We train our model using a 4090 laptop GPU and evaluate its efficiency and other approaches on multiple devices. The size of the images used for training is 256×256 . Because odd resolution is required by DNnet, $n = 3$, we use 255×255 images to train DNnet, $n = 3$. The random horizontal flip probability is 0.5, and learning rate initialized to 0.0005 is adjusted to 0.7 times the original value after every six epochs. We train our model for 30 epochs with a batch size set to 8 using the ADAM optimizer.

To evaluate the performance, we compared our model with eight state-of-the-art learning-based methods, namely, LiteEnhanceNet (Zhang et al., 2024), ASAN (Park & Eom, 2024), DICAM (Tolie, Ren, & Elyan, 2024), FA+Net (Jiang et al., 2023), PUIE (Fu et al., 2022), FUnIE-GAN (Islam et al., 2020), Shallow-UWnet (Naik et al., 2021), and HWCNN (Li et al., 2020) and six learning-free methods, namely, Fusion (Ancuti et al., 2017), RGHS (Huang, Wang, Song, Sequeira, & Mavromatis, 2018), MLE (Zhang, Zhuang, et al., 2022), CLAHE (Zuiderveld, 1994), IBLA (Peng & Cosman, 2017), and GC (Schlick, 1995). All models use the training hyperparameters set in the original paper. Because odd resolutions are necessary to adopt pixelshuffle and pixelunshuffle strategies, DNnet with $n = 3$ uses 255×255 size images for training, and when testing DNnet with $n = 3$ on 720p and 2K image, the resolutions are replaced with 1281×720 and 2562×1440 . Because PUIE requires multiple inferences to generate enhanced images, we only test models that perform one inference to generate results for evaluating the MACs and frames per second (FPS).

4.2. Datasets

(1) The UIEB dataset (Li et al., 2019) contains 890 images with varying degrees of degradation taken from various water environments. Li et al. (2019) generated enhancement results using 12 enhancement methods and selected the best-enhanced image as GT based on an evaluation of 50 volunteers. We randomly selected 800 images from UIEB as the training set and the remaining 90 as the test set.

(2) LSUI (Peng et al., 2023) contains 4279 authentic underwater images of rich water scenes, water types, lighting conditions, and target categories. Peng et al. (2023) scored the enhanced images obtained from 17 UIE methods based on indicators such as underwater image quality measure (UIQM) and UICQE and screened candidate images. The picture with the highest score based on the evaluation of 10 volunteers is selected from the candidate pictures as the reference for the underwater picture. In our experiments, we randomly selected 400 images from 4279 paired images as the test set and the remaining 3879 as the training set.

The resolution of the images in the UIEB and LSUI is not uniform. All images are resized to 256×256 during testing and training unless the odd resolution is required.

(3) The UVEB constructed by Xie et al. (2024), is the first large-scale high-resolution underwater video enhancement benchmark designed to promote the development of underwater vision technologies. The dataset consists of 1308 pairs of video sequences, totaling more than 453,000 high-resolution frames, with 38% of these frames being 4K. UVEB is curated from diverse locations across multiple countries, capturing a wide range of underwater scenes and video degradation types. We randomly extracted some high-resolution underwater images from UVEB. After obtaining multiple pairs of enhanced images and underwater images, we tested these images for feature matching and image segmentation tasks.

Table 4
Quantitative evaluation on the UIEB dataset.

Methods	PSNR(dB)↑	MSE($\times 10^3$)↓	SSIM↑	UIQM↑
GC	16.945	2.027	0.762	2.686
IBLA	17.035	2.633	0.644	1.743
RGHS	24.296	0.452	0.832	2.379
MLLE	18.258	1.320	0.730	2.274
CLAHE	20.748	0.724	0.857	3.087
Fusion	21.503	0.686	0.862	3.128
FUnIE-GAN	20.361	0.750	0.822	3.351
HWCNN	19.799	1.046	0.778	2.682
Shallow-UWnet	19.273	1.006	0.778	2.920
PUIE	24.116	0.385	0.899	3.183
FA+Net	24.530	0.376	0.901	3.167
ASAN	19.878	0.733	0.718	3.135
DICAM	24.966	0.453	0.901	2.979
LiteEnhanceNet	23.186	0.496	0.885	3.127
DNnet, n = 1	26.335	0.368	0.910	2.994
DNnet, n = 2	26.345	0.394	0.901	2.926
DNnet, n = 3	24.892	0.432	0.876	3.014
DNnet, n = 4	25.092	0.457	0.878	2.912

4.3. Quantitative evaluation

4.3.1. Evaluation metrics

To evaluate the quality of generated images, we use four standard evaluation metrics, namely, peak signal-to-noise ratio (PSNR), SSIM, mean squared error (MSE), and no reference metric UIQM.

- (1) MSE (Eq. (16)) measures the pixelwise difference between the enhanced and original images. A lower MSE indicates a closer match and less error. In Eq. (16), N is the number of samples. y_i is the true value of the i th sample. x_i is the predicted value of the i th sample.

$$MSE(x_i, y_i) = \frac{1}{N} \sum_{i=1}^N (x_i - y_i)^2 \quad (16)$$

- (2) PSNR (Eq. (17)) is based on MSE. It measures the ratio between the maximum possible signal and the noise. High PSNR values indicate better image quality and less distortion. In Eq. (17), R is the maximum possible pixel value of the image (e.g., 255 for an 8-bit image).

$$PSNR(x_i, y_i) = 10 \times \log_{10} \left(\frac{R^2}{MSE(x_i, y_i)} \right) \quad (17)$$

- (3) SSIM (Eq. (18)) assesses the similarity in terms of luminance, contrast, and structure between the enhanced and original images. An SSIM value closer to 1 indicates higher similarity and better quality. In Eq. (18), μ_x (μ_y) is the mean of the original (target) images. σ_x^2 (σ_y^2) is the variance of the original (target) images. σ_{xy} is the covariance of the original (target) images. c_1 and c_2 are constants to stabilize the division with weak denominator.

$$SSIM(x, y) = \left(\frac{2\mu_x\mu_y + c_1}{\mu_x^2 + \mu_y^2 + c_1} \right) \left(\frac{2\sigma_{xy} + c_2}{\sigma_x^2 + \sigma_y^2 + c_2} \right) \quad (18)$$

- (4) UIQM (Eq. (19)) is specifically designed for underwater images. It evaluates quality based on certain factors such as color (UICM), contrast (UIConM), and sharpness (UISM). This metric provides a comprehensive measure of enhancement effectiveness according to human visual system stimulation. In Eq. (19), c_1 , c_2 and c_3 are Weights for each component, typically set to 0.028, 0.295, and 3.575 respectively.

$$UIQM = c_1 \times UICM + c_2 \times UIConM + c_3 \times UISM \quad (19)$$

To measure the speed and the efficiency of the methods, we also recorded the parameters, MACs, inference FPS, and memory consumption at 720p, 1080p, 2K, and 4K resolutions. Memory usage is measured

Table 5
Quantitative evaluation on the LSUI dataset.

Methods	PSNR(dB)↑	MSE($\times 10^3$)↓	SSIM↑	UIQM↑
GC	16.637	1.857	0.768	2.789
IBLA	20.916	1.012	0.734	2.239
RGHS	23.692	0.463	0.808	2.472
MLLE	19.055	1.067	0.735	2.388
CLAHE	16.795	1.922	0.700	3.069
Fusion	21.127	0.699	0.816	3.184
FUnIE-GAN	24.840	0.310	0.828	2.975
HWCNN	23.115	0.508	0.852	3.054
Shallow-UWnet	21.444	0.655	0.831	3.060
FA+Net	25.230	0.289	0.872	3.130
ASAN	20.814	0.603	0.778	2.943
DICAM	25.479	0.272	0.865	3.098
LiteEnhanceNet	23.301	0.482	0.840	2.997
DNnet, n = 1	26.041	0.257	0.874	2.910
DNnet, n = 2	25.772	0.265	0.868	2.896
DNnet, n = 3	25.460	0.271	0.859	2.955
DNnet, n = 4	25.527	0.287	0.861	2.845

Table 6

Parameters and MACs when processing images of different resolutions with various methods.

Methods	Parameters(k)↓	MACs(G)↓			
		720p	1080p	2K	4K
FUnIE-GAN	7.02m	143.70	322.75	575.21	1.29T
LiteEnhanceNet	13.69	8.33	18.75	33.33	74.98
Shallow-UWnet	219.46	304.17	684.39	1.22T	2.74T
PUIE ^a	16.12m	—	—	—	—
HWCNN	29.52	27.21	61.22	108.83	244.88
FA+Net	8.99	8.05	18.10	32.21	72.42
ASAN	755.24	686.56	1.54T	2.75T	6.18T
DICAM	943.72	747.75	OOM	OOM	OOM
DNnet, n = 1	3.97	3.66	8.24	14.65	32.96
DNnet, n = 2	15.51	3.57	8.04	14.30	32.17
DNnet, n = 3	34.74	3.56	8.00	14.24	32.02
DNnet, n = 4	61.66	3.55	7.99	14.21	31.97

[OOM] denotes out of 16GB memory.

^a Because PUIE requires multiple inferences to generate enhanced images, we only test models that perform one inference to generate results for evaluating the MACs.

when the image processing speed is stable. In all tables, the upward arrow indicates that when the indicator is large, it is better. The downward arrow indicates that when the indicator is small, it is better. All the best results are marked in red.

4.3.2. Results

Our proposed model contains four versions according to n , which is the scale parameter in pixelshuffle and pixelunshuffle in our model. When the parameter n is large, the size of the feature map processed by the model is small.

Tables 4 and 5 present the results of a quantitative comparison of the UIEB methods. Because the author of PUIE did not provide additional labels for training on the LSUI dataset, the test results for the LSUI dataset exclude PUIE. When pursuing a high quality, we recommend using versions of $n = 1$ or $n = 2$. When $n = 2$, our model achieves the best PSNR score on the UIEB dataset; when $n = 1$, our model achieves the best score in the SSIM and MSE. The UIQM scores of the four models we proposed are relatively high.

Notably, although the feature map size decreases as n increases, the performance of our method does not decrease linearly. We speculate that this phenomenon is attributed to the fact that the PBP structure has the characteristic of increasing more trainable parameters as the feature map size decreases (Table 6). More trainable parameters cannot completely eliminate the negative effect caused by decreased feature map size.

Table 6 presents the parameter quantities of different methods and the MACs at different resolutions. Our proposed method uses the fewest parameters when $n = 1$. When $n = 4$, the MACs of our model are the

Table 7

Memory usage and FPS when processing images of different resolutions with various methods.

Methods	Memory usage(GB)↓				FPS↑			
	720p	1080p	2K	4K	720p	1080p	2K	4K
FUnIE-GAN	1.52	2.98	5.04	10.85	113.25	46.94	25.79	11.02
LiteEnhanceNet	2.40	5.53	9.59	OOM	25.38	11.19	6.30	OOM
Shallow-UWnet	1.40	2.80	4.75	12.33	25.46	11.15	6.08	2.51
PUIE ^a	3.91	8.30	13.41	OOM	–	–	–	–
HWCNN	1.08	2.06	3.45	7.40	96.98	40.02	22.13	9.74
FA+Net	1.25	2.41	3.65	8.25	41.19	19.42	11.43	5.46
ASAN	2.22	4.42	7.55	15.09	3.84	1.70	0.95	0.36
DICAM	14.97	OOM	OOM	OOM	3.11	OOM	OOM	OOM
DNnet, n = 1	1.74	3.37	5.82	12.52	92.45	40.76	22.78	10.08
DNnet, n = 2	0.76	1.17	1.91	3.81	362.58	145.77	77.25	32.86
DNnet, n = 3	0.63	0.82	1.18	2.17	859.85	284.50	138.03	54.37
DNnet, n = 4	0.50	0.69	0.97	1.36	1239.16	515.73	208.55	78.82

[OOM] denotes out of 16GB memory.

^a Because PUIE requires multiple inferences to generate enhanced images, we only test models that perform one inference to generate results for evaluating the FPS.**Table 8**

FPS and Memory usage when our method processes 4K images on NVIDIA Jetson Xavier NX, NVIDIA Jetson AGX Xavier, and NVIDIA Jetson AGX Orin.

Models	FPS↑			Memory usage(GB)↓
	Xavier NX	AGX Xavier	AGX Orin	
DNnet, n = 1	3.036	5.273	12.826	3.000
DNnet, n = 2	10.153	16.132	47.556	1.501
DNnet, n = 3	15.913	26.666	69.847	0.773
DNnet, n = 4	19.226	30.622	78.866	0.669

lowest at 720p, 1080p, 2K, and 4K resolutions. Although the computational cost increases significantly for all tested methods with increasing resolution, our model maintains minimal computational cost. The upward trend of parameters of DNnet with different n mainly follows the analysis of PBP structure discussed in Section 3.1.1. The slight differences in MACs are caused by different sizes of feature maps and biases in convolutional layers.

Table 7 presents the memory usage and FPS processed by different methods when processing images with different resolutions. When n = 4, our method achieves the best results on memory usage and FPS at four resolutions. Using more parameters in exchange for better parallel computational efficiency and faster inference speed, our model can achieve much faster inference speed.

Although this strategy will time the parameters by n^2 , the parameters of our baseline model (DNnet, n = 1) are quite limited. The additional memory usage caused by the increased parameters is significantly less than the memory saved by calculating smaller feature maps. We recommend using DNnet with n = 4 to increase the inference speed and reduce memory usage as much as possible. Additionally, we recommend using DNnet with n = 2 to leverage inference speed and quality.

The adaptability of our model is further demonstrated in our experiments on edge devices, particularly the NVIDIA Jetson Xavier NX, NVIDIA Jetson AGX Xavier, and NVIDIA Jetson AGX Orin (Table 8). We use TensorRT for deployment in FP32 and FP16 mixed precision. Our proposed model can speed up the inference at the expense of using more parameters to improve the parallel computational efficiency. Thus, our proposed DNnet can meet the real-time 4K image enhancement requirements on multiple edge devices.

4.4. Qualitative evaluation

We compare our method with learning-based methods and learning-free methods that perform better in quantitative evaluations in three challenging scenarios in UIE: well-lit scenes, greenish scenes, and bluish scenes. Each scene contains underwater images from the UIEB and LSUI datasets with varying degrees of degradation. Raw represents

unenanced image, and GT represents ground truth.

With artificial lighting and reduced absorption and degradation of sunlight at shallow depths, images captured underwater can be restored to satisfactory quality without much enhancement (Fig. 10). However, UIE methods are prone to overenhance and underenhance underwater images in scenes with good lighting conditions. For instance, the results of Fusion have a noticeable color cast. LiteEnhanceNet and FA+Net tend to make the results appear gray. Although FUnIE-GAN improves the contrast of Fig. 10 raw (a)(b)(c), it adds irregular color spots. RGHS performs better in the blue and green channels but suffers from insufficient enhancement in the red channel. Because we introduce color column loss to address the undersaturation problem, the results obtained using our method in well-lit scenes are closer to GT.

Owing to the presence of phytoplankton, pictures taken in lakes and some sea areas have a greenish cast (Fig. 11 Raw). For learning-free methods, compared with learning-based methods, Fusion and RGHS are unable to generate reasonable results for greenish underwater images with varying degrees of degradation. The saturation of the results obtained using Fusion is significantly low, and Fig. 11 RGHS (a)(c) generated by RGHS have large unnatural blue or green areas. Most of the compared learning-based methods are better at enhancing greenish underwater images with less degradation. However, LiteEnhanceNet and FUnIE-GAN poorly perform in significantly degraded greenish scenes (Fig. 11 Raw (a)(b)(c)). In addition, DICAM and FA+Net cannot correctly enhance the image of scene (Fig. 11 Raw (a)). Owing to the use of the learning-based method to improve compensation, the enhancement results of greenish underwater images with different degradation using our proposed method indicate better contrast brightness and more details.

Because blue light suffers the minor absorption and scattering effect in water, underwater images are often bluish (Fig. 12 Raw). Compared with pictures taken in well-lit scenes and greenish scenes, those taken in bluish scenes lose more color information and are thus more difficult to enhance. Without trainable parameters, learning-free methods such as Fusion and RGHS perform poorly in non-greenish scenes. Significant deviations can be observed in the enhancement results of Fusion and RGHS of degraded bluish images. DICAM and FA+Net do not adequately compensate for the red channel in bluish scenes. Moreover, LiteEnhanceNet and FUnIE-GAN cannot effectively eliminate the bluish problem of the original image. Owing to the entire channel compensation strategy, compared with other methods, our method gets the most satisfactory results in enhancing bluish underwater images with different degrees of degradation.

4.5. Ablation study

This section presents the detailed ablation experiments in terms of three aspects: analysis of parameters n and p, efficiency of modules included in the model, and loss functions.

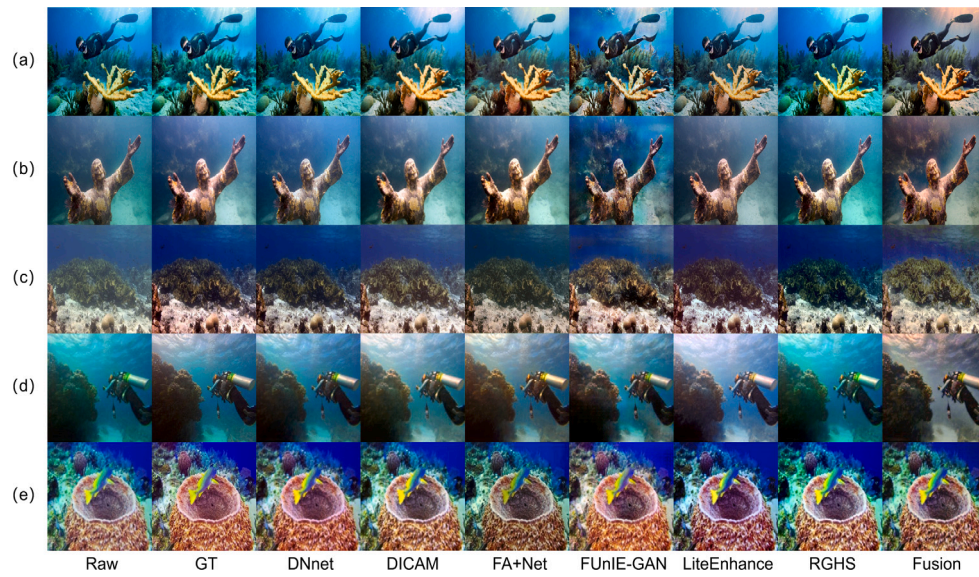


Fig. 10. Enhancement results of underwater images in well-lit scenes on the UIEB and LSUI datasets using different methods.

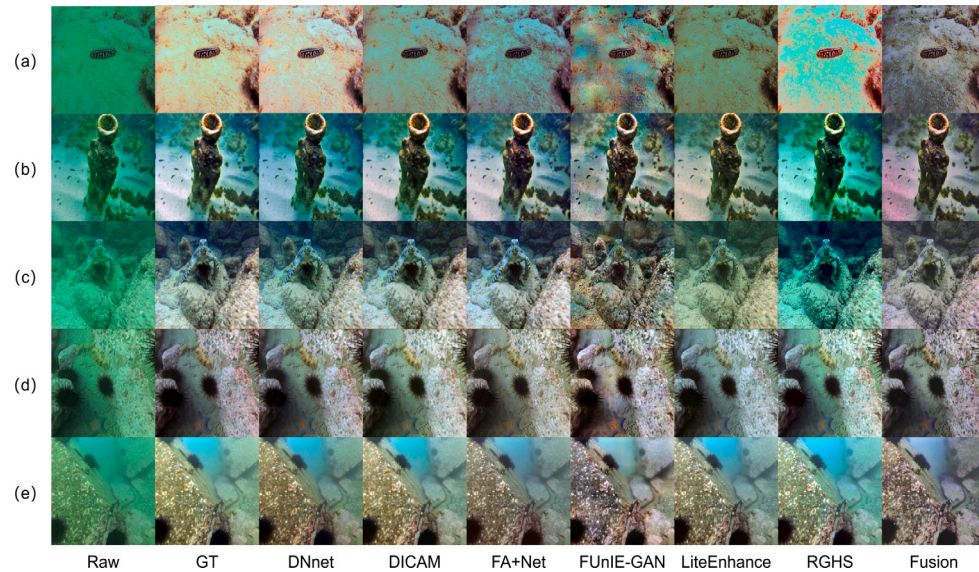


Fig. 11. Enhancement results of underwater images in greenish scenes on the UIEB and LSUI datasets using different methods.

4.5.1. Analysis of different n values

As shown in Tables 4 and 5, when n increases, the SSIM of the results generated by our method decreases and the MSE increases, indicating that the accuracy of the generated results gradually decreases as n increases. When n is equal to 1 or 2, the PSNR of the enhanced image is higher than that when n is equal to 3 or 4. The above phenomena indicate that smaller n values, such as 1 or 2, are more helpful in ensuring enhancement quality than larger n . Fig. 13 presents the enhanced images generated by the model using different n values. When n increases gradually, we notice that the artifacts at the edge of the enhanced image increase gradually as well. Particularly, when $n = 4$, the mosaic is generated in some areas in the generated image. Additionally, a color cast can be observed when $n = 4$. As shown in Tables 6 and 7, we can conclude that the model's running speed can be improved along with the increase in parameter n . However, the quality of the enhanced image decreases when n is too large.

4.5.2. Impact of various p values

As shown in Table 9, when we remove CDR, the model exhibits the worst SSIM and PSNR. When p gradually increases, the PSNR,

Table 9

The effect of different p values on model performance.

Models	UIQM $\uparrow$	SSIM $\uparrow$	PSNR(dB) $\uparrow$	MSE $\downarrow$
w/o CDR ^a	2.996	0.905	25.564	364.105
$p = 1$	3.004	0.908	26.053	370.664
$p = 2$	3.006	0.908	26.074	380.045
$p = 3$	2.991	0.909	26.117	384.687
$p = 4$	2.996	0.909	26.241	380.625
$p = 5$	2.994	0.910	26.335	367.522
$p = 6$	3.003	0.909	26.075	388.037

^a w/o CDR indicates that CDR was removed from the model.

SSIM, and UIQM of the model first show an upward trend before $p = 5$. To further reveal the relationship between DR and PSNR, we investigated the correlation between the PSNR of the image in the UIEB dataset processed by FAN and the DR of the raw image. In Table 10, Kendall's tau-b (Somers, 1962) and Spearman's rho (Costner, 1965)

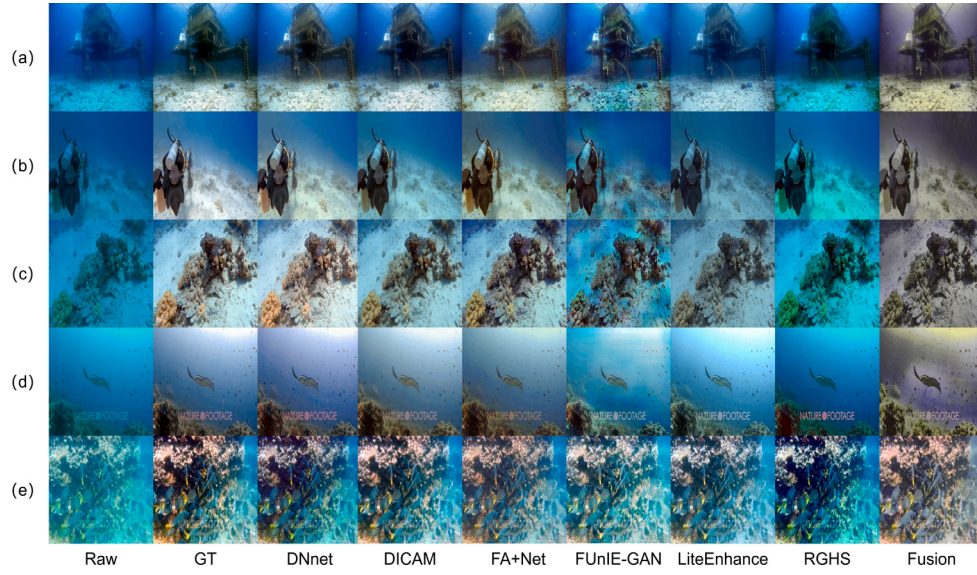


Fig. 12. Enhancement results of blue-toned underwater images on the UIEB and LSUI datasets using different methods.

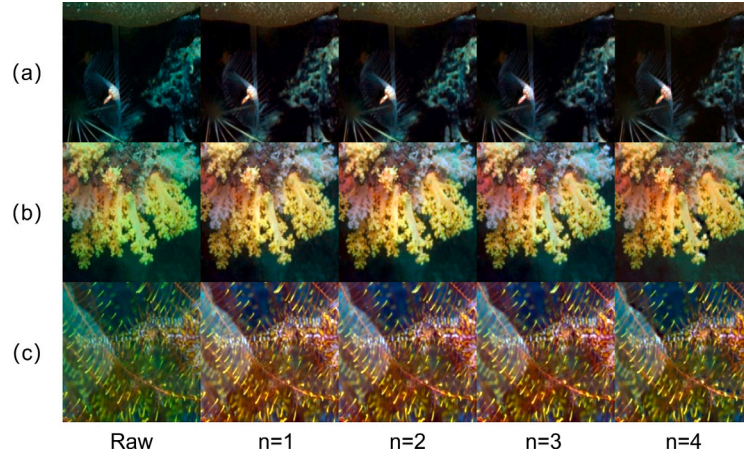


Fig. 13. Enhancement results of DNnet with different n values.

Table 10

Three correlation coefficients between the PSNR of images in the UIEB dataset processed by FAN and DR for the R, G, and B channels.

PSNR and DR	R	G	B
Kendall's tau-b	0.122**	0.117**	0.183**
Spearman's rho	0.182**	0.173**	0.274**
Pearson	0.261**	0.195**	0.280**

** Indicates that the correlation is significant at the 0.01 level (two-tailed).

indicate the correlation of trends between two variables, and Pearson's correlation (Pearson, 1894) indicates the degree of linear correlation between two variables. We found that the DR of the raw image is positively correlated with the PSNR of the image processed by FAN, which means the PSNR of the image processed by FAN is better when the DR of the raw image is large. The results indicate that images with small DR would have a low PSNR after being processed by FAN.

We further explore Pearson coefficients between the PSNR of the image in the UIEB dataset processed by FAN and the C of the raw image to determine how to find a suitable p . According to Table 11, after applying Eq. (14), the Pearson coefficients become more apparent, and the trend is roughly the same as shown in Table 9. These results confirm that DR and C were correlated with image qualities. Specifically,

we design C with a negative correlation with DR , as explained in Section 3.3. Thus, the Pearson coefficients shown in Table 11 are all negative. Notably, most of the absolute values of C shown in Table 11 are larger than the DR in Table 10. The result further verifies that the high-order polynomial function ($p > 1$) is more suitable to describe the relationship between C and the enhanced image quality. We obtain the highest correlation coefficients when $p = 5$.

4.5.3. The impact of each module

Table 12 presents the ablation study of our proposed method after removing CDR and FAN and replacing them with different normalization approaches. Compared with adding CDR, the model without CDR performs slightly better in MSE and UIQM, but worse in SSIM and PSNR. We observe a performance loss compared with the baseline when using batch-wise normalization. Compared with batch-wise normalization, image-wise normalization from FUnIE-GAN (Islam et al., 2020) improves the performance of the baseline model. However, the degradation of light of different wavelengths is inconsistent, and we found that normalization for different color channels indicates better enhancement. A similar structure of channel-wise normalization is apparent in Park and Eom (2024) recent work. By adding FAN, which uses average pooling to reduce overenhancement or underenhancement risks on normalization, the performance of the baseline model significantly improved. With the help of CDR and FAN, the proposed model

Table 11

Pearson coefficient of PSNR of the image in UIEB dataset processed by FAN and C on the R, G, and B channels at different p values.

PSNR and C	p = 1	p = 2	p = 3	p = 4	p = 5	p = 6
R	-0.261**	-0.323**	-0.355**	-0.369**	-0.373**	-0.371**
G	-0.195**	-0.230**	-0.241**	-0.240**	-0.230**	-0.215**
B	-0.280**	-0.317**	-0.328**	-0.326**	-0.316**	-0.302**

** indicates that the correlation is significant at the 0.01 level (two-tailed).

Table 12

Ablation study on each module and different normalization.

Models	UIQM↑	SSIM↑	PSNR(dB)↑	MSE↓
baseline (w/o CDR, FAN ^a)	2.922	0.843	20.362	956.586
w/ FAN ^b	2.996	0.905	25.564	364.105
w/ channel-wise normalization	3.077	0.896	24.240	435.516
w/ image-wise normalization	3.008	0.869	22.088	573.955
w/ batch-wise normalization	2.500	0.641	18.696	1085.183
full model	2.994	0.910	26.335	367.522

^a w/o CDR indicates that CDR was removed from the model.

^b w/ FAN means adding FAN to our model and removing CDR.

Table 13

Results obtained by training using different combinations of loss functions.

Loss ^a	PSNR(dB)↑	MSE($\times 10^3$)↓	SSIM↑	UIQM↑
Full loss	26.335	0.368	0.910	2.994
SSIM loss only	25.640	0.395	0.912	3.032
Color column loss only	24.106	0.401	0.723	2.919

^a Each combination has the same coefficients as those given in Eq. (15).

(full model) achieves the best performance.

4.5.4. Contributions of each loss terms

Table 13 presents the results obtained by training the model using various combinations of loss functions. When using the loss function combination of Eq. (15), the MSE and UIQM indicators of the model enhancement result are the best. When the model is trained only with the color column loss, the generated results perform worse than those obtained by our proposed method in terms of PSNR, MSE, SSIM, and UIQM. SSIM loss directly utilizes the SSIM metric as the loss function, allowing the model to achieve the best SSIM when trained with SSIM loss alone. Since color column loss is not a pixel-level loss and is only calculated over a few intervals with significant color distribution differences, relying solely on color column loss does not provide a substantial improvement in metrics. However, we observe that when SSIM loss is combined with color column loss, there is a notable improvement in PSNR and MSE. This indicates that training with the full loss function helps preserve more image details and accuracy, especially for subtle differences. Considering that this combination can ensure more details beneficial to back-end tasks, we ultimately opt for combining color column loss and SSIM loss for training our model.

4.6. The significance of enhancing high-resolution underwater images

We present a comprehensive set of experiments that demonstrate the advantages of processing underwater images at higher resolutions, particularly in the context of image segmentation and feature matching tasks.

In our experiments, we evaluated 4K images from UVEB (Xie et al., 2024) dataset at multiple resolution levels, ranging from 720p (1280 $\times$ 720) to 4K (3840 $\times$ 2160). All enhanced images of different resolutions are obtained by inputting the raw underwater images of corresponding resolutions into DNnet (n = 1) trained with UIEB (Li et al., 2019). For each resolution, we measure the number of objects in segmentation or correct matches in feature matching to demonstrate

Table 14

The comparison of the segmentation results.

Enhancement	720p	1080p	2K	4K
Before	106	116	110	113
After	121	139	138	152

Table 15

The comparison of the feature matching results, where the enhanced 4K resolution images consistently outperformed lower resolutions images and raw images in the number of correct matches.

Enhancement	720p	1080p	2K	4K
Before	1981	5561	9819	10587
After	3549	9122	16347	20903

the performance.

Our experiments (Sections 4.6.1 and 4.6.2) validate that even if the human eye may not be able to fully perceive the differences between high-resolution images at a glance, the enhanced high-resolution underwater image contributes significantly to performance in various computer vision tasks.

4.6.1. Advantages of high-resolution UIE in segmentation

Image segmentation is a critical step in many computer vision tasks, where the objective is to partition an image into meaningful regions or objects (Zhang, Zhao, Zhang, Peng, & Fan, 2022). We use the default SAM (Kirillov et al., 2023) with ViT-large (Dosovitskiy, 2020) as predictor to complete the segmentation task. The results shown in Table 14 demonstrate that higher-resolution UIE improved searching ability especially in the presence of small or intricate structures that would otherwise be blurred or merged at lower resolutions. In addition, as shown in Fig. 14, we also found in that high-resolution UIE helps refine the boundaries of segmentation.

4.6.2. Advantages of high-resolution UIE in feature matching

Feature matching, which involves identifying corresponding points or regions between different images, benefits significantly from high-resolution input (Yang, Manela, Happold, & Ramanan, 2019). In our feature matching experiments, we tested the SIFT (Chiu, Chang, Chen, & Chang, 2013) descriptors at different resolutions, and applied brute force (Heule & Kullmann, 2017) algorithm on pairs of images in UVEB (Xie et al., 2024). As shown in Table 15, higher resolution contributes to a significant increase in the number of matched points. At the same resolution, the number of correct matches in enhanced underwater images is nearly double that of raw images (Fig. 15). Based on these results, we conclude that high-resolution UIE enhances key point localization, thereby improving the performance of the feature matching algorithm.

5. Conclusion

We propose DNnet with a lightweight structure that can handle a 4K UIE task in real time on multiple edge devices. We achieve a

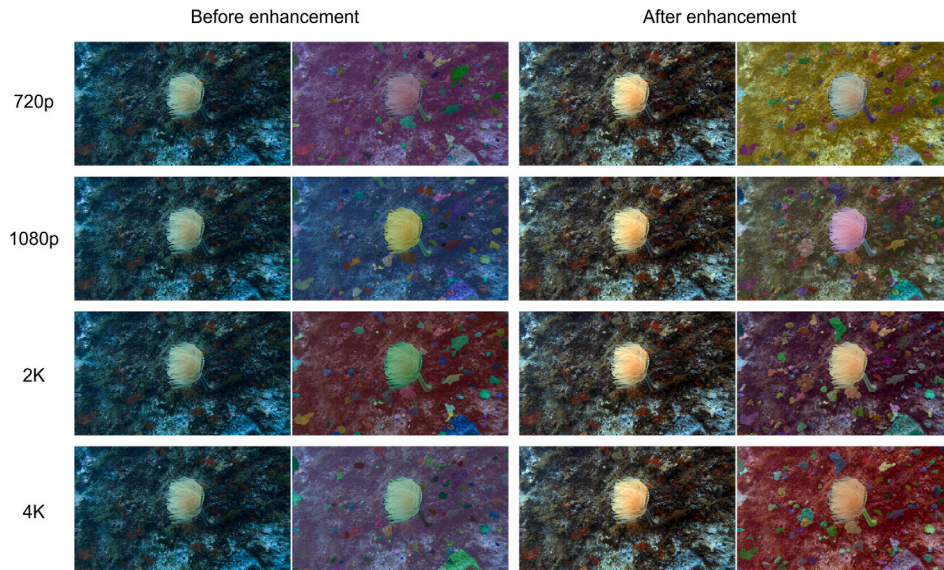


Fig. 14. Comparison of segmentation results between enhanced underwater images and raw underwater images at different resolutions.

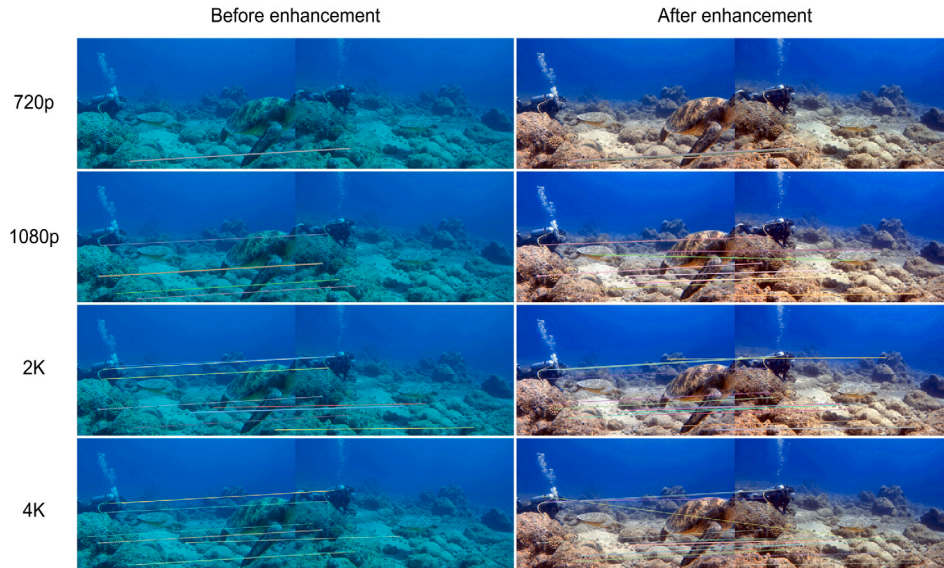


Fig. 15. Comparison of matching results between enhanced underwater images and raw underwater images at different resolutions.

fast inference speed using PBP designed according to the strategy of improving the parallel computational efficiency at the cost of more parameters. Our proposed FAN module that aimed to mitigate the impact of overly bright or dark areas in underwater images demonstrated better robustness than other normalization methods. The proposed CDR that captures the relationship between DR and enhancement quality improves performance by coordinating PBP and FAN. Additionally, our formulated color column loss that focuses on the difference in local color distribution between undersaturated images and GT improves training results. Therefore, compared with other state-of-the-art methods, our proposed method achieves the best performance while exhibiting the fastest inference speed.

Underwater video enhancement is a more challenging work than an UIE task. Because two adjacent frames can preserve abundant information, this abundant information could be used for extra restrictions to guide an enhancement training process. Our future work will focus on utilizing the abundant information between adjacent frames and explore the possibility of real-time underwater video enhancement.

CRediT authorship contribution statement

Tianyu Cao: Conceptualization, Methodology, Software, Writing – Original Draft, Writing – Review & Editing, Investigation. **Zhibin Yu:** Writing – Review & Editing, Supervision, Project administration. **Bing Zheng:** Formal analysis, Writing – Review & Editing, Validation, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Ancuti, C. O., Ancuti, C., De Vleeschouwer, C., & Bekaert, P. (2017). Color balance and fusion for underwater image enhancement. *IEEE Transactions on Image Processing*, 27(1), 379–393. <http://dx.doi.org/10.1109/TIP.2017.2759252>.
- Aydin, T. O., Mantiuk, R., Myszkowski, K., & Seidel, H.-P. (2008). Dynamic range independent image quality assessment. *ACM Transactions on Graphics*, 27(3), 1–10. <http://dx.doi.org/10.1145/1360612.1360668>.
- Chang, B., Li, J., Ren, L., & Chen, Z. (2024). Dual branch Transformer-CNN parametric filtering network for underwater image enhancement. *Journal of Visual Communication and Image Representation*, 100, <http://dx.doi.org/10.1016/j.jvcir.2024.104131>.
- Chaurasiya, R. K., & Ramakrishnan, K. (2013). High dynamic range imaging. In *2013 international conference on communication systems and network technologies* (pp. 83–89). <http://dx.doi.org/10.1109/CSNT.2013.27>.
- Chen, F., Li, S., Han, J., Ren, F., & Yang, Z. (2023). Review of lightweight deep convolutional neural networks. *Archives of Computational Methods in Engineering*, 1–23. <http://dx.doi.org/10.1007/s11831-023-10032-z>.
- Chiu, L. C., Chang, T. S., Chen, J. Y., & Chang, Y. C. (2013). Fast SIFT design for real-time visual feature extraction. *IEEE Transactions on Image Processing*, 22(8), 3158–3167. <http://dx.doi.org/10.1109/TIP.2013.2259841>.
- Costner, H. L. (1965). Criteria for measures of association. *American Sociological Review*, 341–353. <http://dx.doi.org/10.2307/2090715>.
- Dosovitskiy, A. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. arXiv preprint [arXiv:2010.11929](https://arxiv.org/abs/2010.11929).
- Fu, Z., Wang, W., Huang, Y., Ding, X., & Ma, K.-K. (2022). Uncertainty inspired underwater image enhancement. In *European conference on computer vision* (pp. 465–482). Springer, http://dx.doi.org/10.1007/978-3-031-19797-0_27.
- Guan, Y., Liu, X., Yu, Z., Wang, Y., Zheng, X., Zhang, S., et al. (2023). Fast underwater image enhancement based on a generative adversarial framework. *Frontiers in Marine Science*, 9, Article 964600. <http://dx.doi.org/10.3389/fmars.2022.964600>.
- Guo, C., Wu, R., Jin, X., Han, L., Zhang, W., Chai, Z., et al. (2023). Underwater ranker: Learn which is better and how to be better. In *Proceedings of the AAAI conference on artificial intelligence*, vol. 37 (pp. 702–709). <http://dx.doi.org/10.1609/aaai.v37i1.25147>.
- Hao, Z., Longxiang, G., Xiangchun, L., Fei, L., & Jiawei, Y. (2023). An underwater imaging method of enhancement via multi-scale weighted fusion. *Frontiers in Marine Science*, <http://dx.doi.org/10.3389/fmars.2023.1150593>.
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770–778).
- Heule, M. J., & Kullmann, O. (2017). The science of brute force. *Communications of the ACM*, 60(8), 70–79. <http://dx.doi.org/10.1145/3107239>.
- Hitam, M. S., Awalludin, E. A., Yusof, W. N. J. H. W., & Bachok, Z. (2013). Mixture contrast limited adaptive histogram equalization for underwater image enhancement. In *2013 international conference on computer applications technology* (pp. 1–5). IEEE, <http://dx.doi.org/10.1109/ICCAT.2013.6522017>.
- Howard, A. G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., et al. (2017). Mobilenets: Efficient convolutional neural networks for mobile vision applications. <http://dx.doi.org/10.48550/arXiv.1704.04861>, arXiv preprint [arXiv:1704.04861](https://arxiv.org/abs/1704.04861).
- Hu, J., Shen, L., & Sun, G. (2018). Squeeze-and-excitation networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 7132–7141).
- Huang, X., & Belongie, S. (2017). Arbitrary style transfer in real-time with adaptive instance normalization. In *Proceedings of the IEEE international conference on computer vision* (pp. 1501–1510).
- Huang, D., Wang, Y., Song, W., Sequeira, J., & Mavromatis, S. (2018). Shallow-water image enhancement using relative global histogram stretching based on adaptive parameter acquisition. In *MultiMedia modeling: 24th international conference, MMM 2018, Bangkok, Thailand, February 5–7, 2018, proceedings, part I 24* (pp. 453–465). Springer, http://dx.doi.org/10.1007/978-3-319-73603-7_37.
- Islam, M. J., Xia, Y., & Sattar, J. (2020). Fast underwater image enhancement for improved visual perception. *IEEE Robotics and Automation Letters*, 5(2), 3227–3234. <http://dx.doi.org/10.1109/LRA.2020.2974710>.
- Jha, M., & Bhandari, A. K. (2024). CBLA: Color balanced locally adjustable underwater image enhancement. *IEEE Transactions on Instrumentation and Measurement*, <http://dx.doi.org/10.1109/tim.2024.3396850>.
- Jiang, J., Ye, T., Bai, J., Chen, S., Chai, W., Jun, S., et al. (2023). Five A + network: You only need 9K parameters for underwater image enhancement. arXiv preprint [arXiv:2305.08824](https://arxiv.org/abs/2305.08824).
- Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., et al. (2023). Segment anything. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 4015–4026).
- Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. *Advances in Neural Information Processing Systems*, 25, <http://dx.doi.org/10.1145/3065386>.
- Lee, Y., Hwang, J.-w., Lee, S., Bae, Y., & Park, J. (2019). An energy and GPU-computation efficient backbone network for real-time object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops*.
- Li, C., Anwar, S., & Porikli, F. (2020). Underwater scene prior inspired deep underwater image and video enhancement. *Pattern Recognition*, 98, Article 107038. <http://dx.doi.org/10.1016/j.patcog.2019.107038>.
- Li, J., Fang, F., Mei, K., & Zhang, G. (2018). Multi-scale residual network for image super-resolution. In *Proceedings of the European conference on computer vision* (pp. 517–532).
- Li, C., Guo, C., Ren, W., Cong, R., Hou, J., Kwong, S., et al. (2019). An underwater image enhancement benchmark dataset and beyond. *IEEE Transactions on Image Processing*, 29, 4376–4389. <http://dx.doi.org/10.1109/TIP.2019.2955241>.
- Liang, J., Cao, J., Sun, G., Zhang, K., Van Gool, L., & Timofte, R. (2021). Swinir: Image restoration using swin transformer. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 1833–1844).
- Liang, Z., Wang, Y., Ding, X., Mi, Z., & Fu, X. (2021). Single underwater image enhancement by attenuation map guided color correction and detail preserved dehazing. *Neurocomputing*, 425, 160–172. <http://dx.doi.org/10.1016/j.neucom.2020.03.091>.
- Ma, X., Dai, X., Bai, Y., Wang, Y., & Fu, Y. (2024). Rewrite the stars. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 5694–5703).
- Muniraj, M., & Dhandapani, V. (2021). Underwater image enhancement by combining color constancy and dehazing based on depth estimation. *Neurocomputing*, 460, 211–230. <http://dx.doi.org/10.1016/j.neucom.2021.07.003>.
- Naik, A., Swarnakar, A., & Mittal, K. (2021). Shallow-uwnet: Compressed model for underwater image enhancement (student abstract). In *Proceedings of the AAAI conference on artificial intelligence*, vol. 35 (pp. 15853–15854). <http://dx.doi.org/10.1609/aaai.v35i18.17923>.
- Pandey, M., Fernandez, M., Gentile, F., Isayev, O., Tropsha, A., Stern, A. C., et al. (2022). The transformational role of GPU computing and deep learning in drug discovery. *Nature Machine Intelligence*, 4(3), 211–221. <http://dx.doi.org/10.1038/s42256-022-00463-x>.
- Park, C. W., & Eom, I. K. (2024). Underwater image enhancement using adaptive standardization and normalization networks. *Engineering Applications of Artificial Intelligence*, 127, Article 107445. <http://dx.doi.org/10.1016/j.engappai.2023.107445>.
- Pearson, K. (1894). Contributions to the mathematical theory of evolution. *Philosophical Transactions of the Royal Society of London. A*, 185, 71–110. <http://dx.doi.org/10.1098/rsp1893.0079>.
- Peng, Y.-T., & Cosman, P. C. (2017). Underwater image restoration based on image blurriness and light absorption. *IEEE Transactions on Image Processing*, 26(4), 1579–1594. <http://dx.doi.org/10.1109/tip.2017.2663846>.
- Peng, L., Zhu, C., & Bian, L. (2023). U-shape transformer for underwater image enhancement. *IEEE Transactions on Image Processing*, <http://dx.doi.org/10.1109/tip.2023.3276332>.
- Pizer, S. M., Amburn, E. P., Austin, J. D., Cromartie, R., Geselowitz, A., Greer, T., et al. (1987). Adaptive histogram equalization and its variations. *Computer Vision, Graphics, and Image Processing*, 39(3), 355–368. [http://dx.doi.org/10.1016/S0734-189X\(87\)80186-X](http://dx.doi.org/10.1016/S0734-189X(87)80186-X).
- Raveendran, S., Patil, M. D., & Birajdar, G. K. (2021). Underwater image enhancement: a comprehensive review, recent trends, challenges and applications. *Artificial Intelligence Review*, (1), <http://dx.doi.org/10.1007/s10462-021-10025-z>.
- Rezende, D. J., Mohamed, S., & Wierstra, D. (2014). Stochastic backpropagation and approximate inference in deep generative models. In *International conference on machine learning* (pp. 1278–1286). PMLR.
- Salman, A., Jalal, A., Shafait, F., Mian, A., Shortis, M., Seager, J., et al. (2016). Fish species classification in unconstrained underwater environments based on deep learning. *Limnology and Oceanography: Methods*, 14(9), 570–585. <http://dx.doi.org/10.1002/lom3.10113>.
- Schlick, C. (1995). Quantization techniques for visualization of high dynamic range pictures. In *Photorealistic rendering techniques* (pp. 7–20). Springer, http://dx.doi.org/10.1007/978-3-642-87825-1_2.
- Shi, W., Caballero, J., Huszar, F., Totz, J., Aitken, A. P., Bishop, R., et al. (2016). Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 1874–1883).
- Somers, R. H. (1962). A new asymmetric measure of association for ordinal variables. *American Sociological Review*, 799–811. <http://dx.doi.org/10.2307/2090408>.
- Tang, Y., Iwaguchi, T., Kawasaki, H., Sagawa, R., & Furukawa, R. (2022). AutoEnhancer: Transformer on U-net architecture search for underwater image enhancement. In *Proceedings of the Asian conference on computer vision* (pp. 1403–1420).
- Tolie, H. F., Ren, J., & Elyan, E. (2024). DICAM: Deep inception and channel-wise attention modules for underwater image enhancement. *Neurocomputing*, 584, Article 127585. <http://dx.doi.org/10.1016/j.neucom.2024.127585>.
- Treibitz, T., & Schechner, Y. Y. (2009). Active polarization descattering. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 31, 385–399. <http://dx.doi.org/10.1109/TPAMI.2008.85>.
- Wang, Z., Bovik, A. C., Sheikh, H. R., & Simoncelli, E. P. (2004). Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4), 600–612. <http://dx.doi.org/10.1109/tip.2003.819861>.

- Xie, Y., Kong, L., Chen, K., Zheng, Z., Yu, X., Yu, Z., et al. (2024). UVEB: A large-scale benchmark and baseline towards real-world underwater video enhancement. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 22358–22367).
- Yan, X., Qin, W., Wang, Y., Wang, G., & Fu, X. (2022). Attention-guided dynamic multi-branch neural network for underwater image enhancement. *Knowledge-Based Systems*, 258, Article 110041. <http://dx.doi.org/10.1016/j.knosys.2022.110041>.
- Yang, G., Manela, J., Hapgood, M., & Ramanan, D. (2019). Hierarchical deep stereo matching on high-resolution images. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*.
- Yeh, C. H., Lin, C. H., Kang, L. W., Huang, C. H., & Wang, C. C. (2021). Lightweight deep neural network for joint learning of underwater object detection and color conversion. *IEEE Transactions on Neural Networks and Learning Systems*, PP(99), <http://dx.doi.org/10.1109/TNNLS.2021.3072414>.
- Yu, F., & Koltun, V. (2015). Multi-scale context aggregation by dilated convolutions. <http://dx.doi.org/10.48550/arXiv.1511.07122>, arXiv preprint [arXiv:1511.07122](https://arxiv.org/abs/1511.07122).
- Zhang, D., Wu, C., Zhou, J., Zhang, W., Li, C., & Lin, Z. (2023). Hierarchical attention aggregation with multi-resolution feature learning for GAN-based underwater image enhancement. *Engineering Applications of Artificial Intelligence*, 125, Article 106743. <http://dx.doi.org/10.1016/j.engappai.2023.106743>.
- Zhang, S., Zhao, S., An, D., Li, D., & Zhao, R. (2024). LiteEnhanceNet: A lightweight network for real-time single underwater image enhancement. *Expert Systems with Applications*, 240, Article 122546. <http://dx.doi.org/10.1016/j.eswa.2023.122546>.
- Zhang, X., Zhao, W., Zhang, W., Peng, J., & Fan, J. (2022). Guided filter network for semantic image segmentation. *IEEE Transactions on Image Processing*, 31, 2695–2709. <http://dx.doi.org/10.1109/TIP.2022.3160399>.
- Zhang, X., Zhou, X., Lin, M., & Sun, J. (2018). Shufflenet: An extremely efficient convolutional neural network for mobile devices. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 6848–6856).
- Zhang, W., Zhuang, P., Sun, H.-H., Li, G., Kwong, S., & Li, C. (2022). Underwater image enhancement via minimal color loss and locally adaptive contrast enhancement. *IEEE Transactions on Image Processing*, 31, 3997–4010. <http://dx.doi.org/10.1109/tip.2022.3177129>.
- Zheng, M., & Luo, W. (2022). Underwater image enhancement using improved CNN based defogging. *Electronics*, 11(1), 150. <http://dx.doi.org/10.3390/electronics11010150>.
- Zhou, J., Pang, L., Zhang, D., & Zhang, W. (2023). Underwater image enhancement method via multi-interval subhistogram perspective equalization. *IEEE Journal of Oceanic Engineering*, <http://dx.doi.org/10.1109/JOE.2022.3223733>.
- Zhuang, P., Li, C., & Wu, J. (2021). Bayesian retinex underwater image enhancement. *Engineering Applications of Artificial Intelligence*, 101, Article 104171. <http://dx.doi.org/10.1016/j.engappai.2021.104171>.
- Zoph, B., & Le, Q. V. (2016). Neural architecture search with reinforcement learning. <http://dx.doi.org/10.48550/arXiv.1611.01578>, arXiv preprint [arXiv:1611.01578](https://arxiv.org/abs/1611.01578).
- Zuiderveld, K. (1994). Contrast limited adaptive histogram equalization. In *Graphics gems IV* (pp. 474–485). <http://dx.doi.org/10.5555/180895.180940>.